

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 3rd Year 1st Term Examination, 2016.
CE 3107
(Elements of Solid Mechanics)

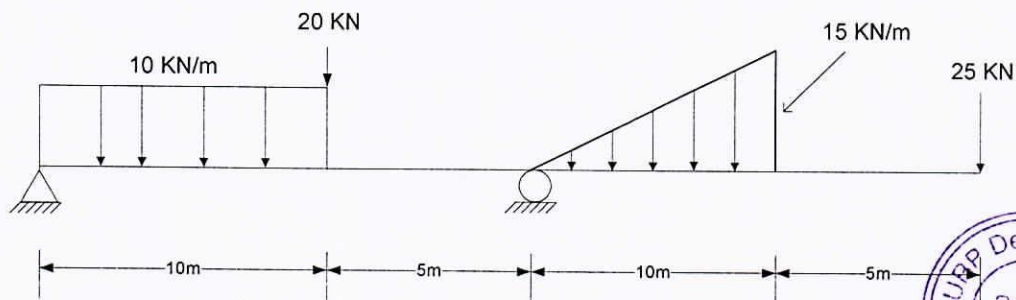
Full Marks: 210

Time: 3 hrs

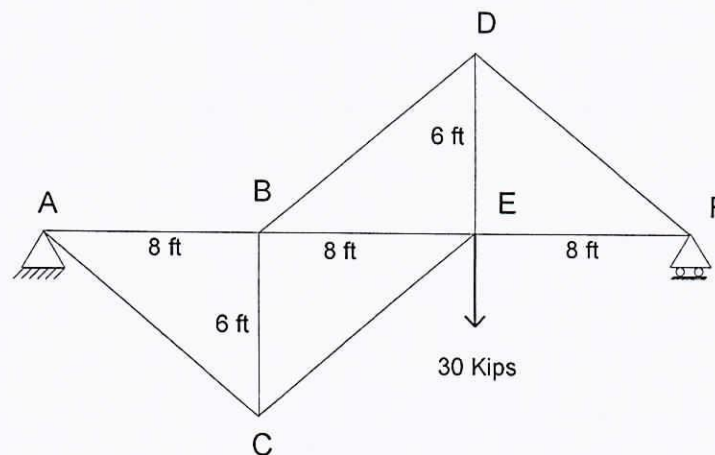
- N.B.** i) Figures in the right margin indicate full marks.
iii) Answer any **three** questions from each section in separate script.

Section – A

1. a. Define i) Shear Force, ii) Bending Moment, iii) Inflection Point, iv) Dangerous Section. (12)
- b. Show the relationship between load, shear and moment. (08)
- c. Draw the shear and bending moment diagrams for the beam shown in figure (15) below-



2. a. What is Poisson ratio? Explain with example. (07)
- b. Show that a uniform stress distribution can exist only if the resultant of the applied loads passes through the centroid of the cross-section. (10)
- c. For the truss shown in figure calculate the stresses in members CE, DE and DF. (18)
The cross-sectional area of each member is 1.8 in^2 . Indicate tension (T) or compression (C)

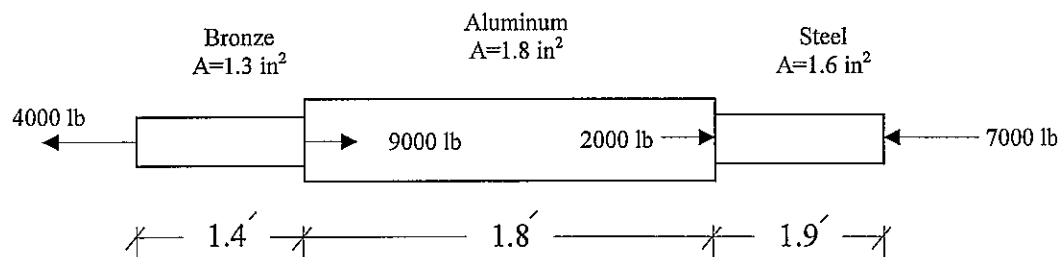


3. a. Define the following terms: i) Factor of safety, ii) Modulus of resilience, iii) Modulus of toughness, iv) Strain hardening, v) Tangent modulus. (15)
- b. Draw the stress-strain diagram of steel and explain its different features. (07)

- c. Following observations were made during the tensile test of a mild steel specimen. (13)
 Diameter of the specimen=50mm
 Length of the specimen=200mm
 Extension under a load of 40KN= 333×10^{-4} mm (Within proportional limit)
 Load at yield point=154.5 KN
 Maximum load=270KN
 Length of the specimen after fracture=252.4mm

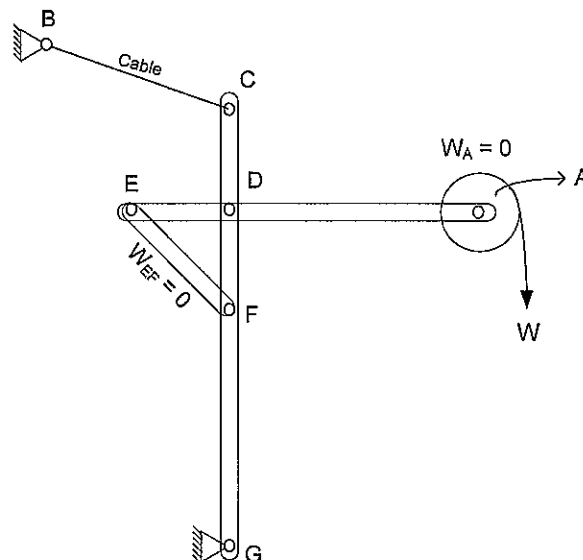
Calculate the values of - i) E, ii) Yield point stress, iii) Ultimate stress and iv) Percentage elongation

4. a. Point out the assumption made for deriving the relation between bending moment and flexure stress. (07)
 b. Obtain an expression for the elongation δ of a vertical bar of length L, uniform cross-sectional area of A, supported at its upper ends under its own weight W. (11)
 c. A composite bar consists of an aluminum section rigidly fastened between a bronze section and a steel section as shown in figure below. Axial loads are applied at positions indicated. Determine the stress in each section. (17)

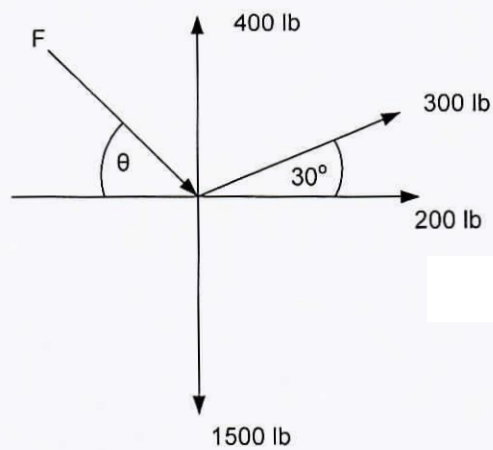


Section – B

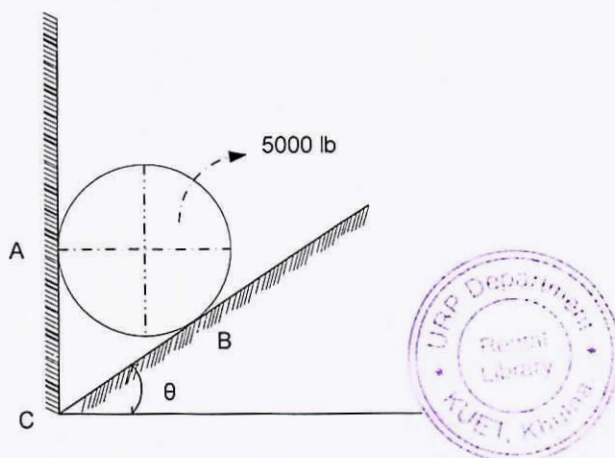
5. a. Define – i) Transmissibility of Force, ii) Centroid, iii) Force, iii) Moment of Inertia (12)
 b. Draw the free body diagram of each member of the following figure. (12)



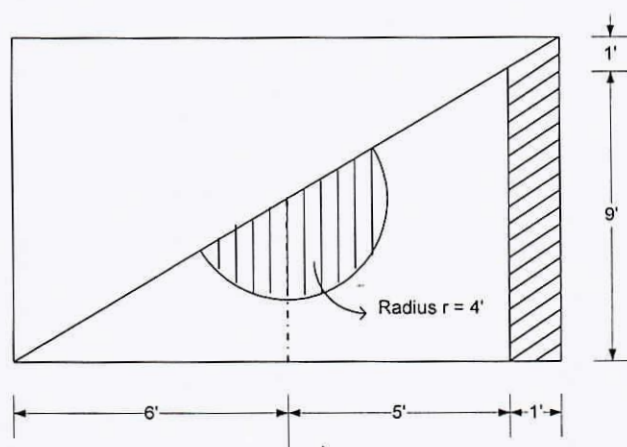
- c. Find the value of F and θ of the following force system shown in figure below. (11)
 Also find the resultant.



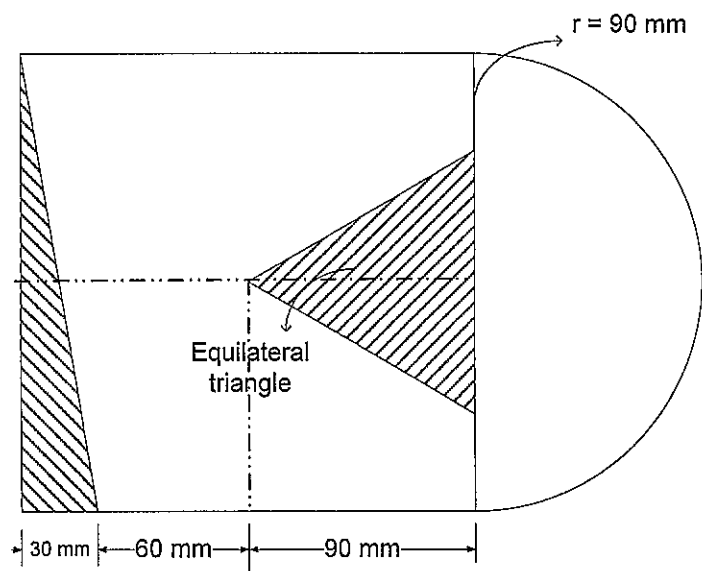
6. a. Differentiate between collinear force system and concurrent force system. (05)
- b. A 5000lb sphere rests on a smooth plane inclined at an angle $\theta=45^\circ$ with the horizontal and against a smooth vertical wall. What are the reactions at the contact surfaces A and B? (15)



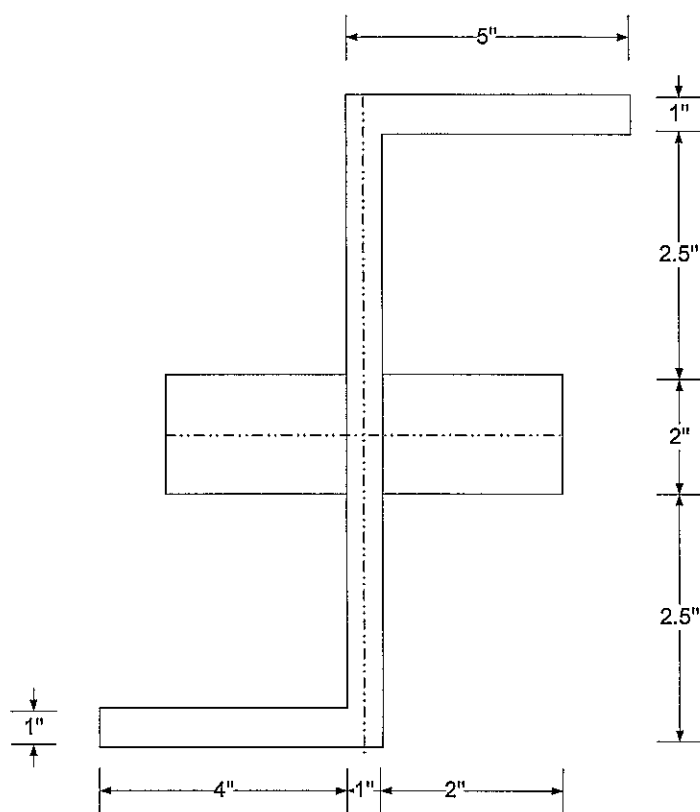
- c. Determine $\bar{x}$ for the unshaded area as shown in the figure below. (15)



7. a. Find the location of the centroid for the area enclosed by the arcs of the two parabolas $y^2 = 9x$ and $x^2 = 4y$, where the linear units are in meters. (17)
- b. Find the moment of inertia of the unshaded area shown in the figure about its base and centroidal axis. (18)



8. a. Find the centroid of a right circular cone, whose altitude is h and whose base has a radius of r . (10)
- b. Derive the expression for the moment of inertia of a sphere about a diameter. The density of the material is constant. (10)
- c. Determine the maximum and minimum moment of inertia of the following figure. (15)



Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 3rd Year 1st Term Examination, 2016.
CE 3151
(Fundamentals of Transportation Engineering)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define transportation engineering. Discuss the advantages and disadvantages of water and rail transport. (08)
(b) What are the diverse characteristics of transportation? Describe the process of transportation yield management. (09)
(c) Classify roads according to their functions and explain them briefly. (09)
(d) What are the objectives of O-D survey? Describe the common methods for collecting O-D data. (09)
2. (a) Define transportation planning. What are the elements of effective transportation planning? (09)
(b) Explain the four stages of transportation planning. (10)
(c) What do you mean by level of service? What are the factors that affecting the level of service? (09)
(d) What are the difficulties of transportation planning process in Bangladesh? (07)
3. (a) What do you mean by traffic engineering? Why traffic engineering is so essential for present situation of Bangladesh? (07)
(b) Write down the differences between the following terms: (09)
(i) Shoulder and Footpath, (ii) Intersection and Interchange, (iii) Volume and Capacity.
(c) Describe why PIEV time is required to determine? Explain the factors that affect the road use characteristics. (10)
(d) Explain the relationship between transportation demand and supply with example. (09)
4. (a) Define the following terms: (08)
(i) Parking load, (ii) Parking Index, (iii) ADT, (iv) AADT
(b) Describe parallel and angle type of parking. Which type of parking do you prefer for what type of situation? (10)
(c) Explain the four stages of transportation modeling. (10)
(d) Discuss the factors affecting traffic volume. (07)



Section – B

5. (a) Define camber. Why is it provided in road surface? What are the factors which affect amount of camber? (08)
- (b) Explain terms:
i) Kerbs, ii) Formation width, iii) Right of way, iv) Friction, v) Traffic separators (10)
- (c) Differentiate between skid and slip. (04)
- (d) The speeds of overtaking and overtaken vehicles are respectively 80 km/hr and 45 km/hr. If the acceleration of the overtaking vehicle is 3 km/hr per second, determine the safe passing sight distance for both one way and two way traffic. (13)
6. (a) Define sight distance and explain the factors causing restrictions in sight distance. Explain the terms reaction time and braking time. (12)
- (b) Derive an expression for finding stopping sight distance at level road. (12)
- (c) The design speed of a highway is 80 kmph. There is a horizontal curve of radius 140 km. Calculate the super elevation needed to maintain this speed. If maximum super elevation of 1 in 15 is not to be exceeded, calculate to the maximum allowable speed on this horizontal curve. (11)
7. (a) With the help of neat sketches, explain the methods of introducing super elevation. (09)
- (b) Why transition curves are introduced at the beginning and end of the sharp horizontal curves? Explain conditions of formation of both summit curve and valley curves. (09)
- (c) What do you mean by ruling gradient and grade compensation? (05)
- (d) A valley curve is divided by descending grade of 1 in 30 meeting an ascending grade of 1 in 25. Design the length of valley curve to fulfill both comfort condition and head light sight distance requirement for a design speed of 80 kmph. Assume allowable rate of change of centrifugal acceleration. $C=0.6 \text{ m/sec}^3$ (12)
8. (a) What do you mean by traffic management and what are the objectives of it? (06)
- (b) What are the consequences of conflicts in an intersection? What are the most effective areas for one way traffic operation? What are the negative consequences of one way operation? (09)
- (c) Draw the potential conflicting points at intersection with no priority for two way, partial one way and full one way system. (12)
- (d) Why is pedestrian facility required? What are the provisions for elderly and handicapped pedestrians? (08)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2016
HUM 3117
(Accounting and Auditing for Planners)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script including **question no. 4 from section A.**
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define accounting (05)
(b) What are the features of transactions? (10)
(c) Distinction between journal and ledger. (10)
(d) State the objectives of trial balance. (10)

2. (a) Affleck's repair shop was started on May 1 by B. Affleck. A summary of May transactions is presented below: (28)

2015 May 01	Invested \$10,000 cash to start the repair shop.
2015 May 02	Purchased equipment for \$5,000 cash.
2015 May 03	Paid \$400 cash for May office rent.
2015 May 04	Paid \$500 cash for supplies.
2015 May 05	Incurred \$250 of advertising costs in the Beacon News on account.
2015 May 06	Received \$4,100 in cash from customers for repair service.
2015 May 07	Withdrew \$ 500 for personal use.
2015 May 08	Paid employee salaries \$ 1000.
2015 May 09	Paid utility bill \$ 140.
2015 May 10	Provided repair service on account to customers \$850.
2015 May 11	Collected cash of \$120 for services billed in transaction (10).

Instruction: Journalize the above transactions in the book of B. Affleck.

(b) What are the elements of accounting equation? (07)

3. (a) Transaction of Hakim Co. for the month of May are given below. (30)

May-1 2016	Started an office as a financial advisor investing \$7500 in cash.
May-5	Pays insurance in advance for the year \$2400 cash.
May-8	Received cash of \$2500 from clients for service rendered.
May-10	Paid secretary salary in cash \$800
May-12	Withdrew cash \$500 for personal use.

Instruction: Prepare necessary ledger accounts.

(b) What are the limitations of a Trial Balance? (05)

4. Global Graphics Company was organized on January 1, 2016, by Jill Jay. At the end of the first 6 months of operations, the trial balance contained the following accounts. (35)

Account Titles	Debits (\$)	Account Titles	Credits (\$)
Cash	9500	Notes Payable	17000
Accounts Receivable	14000	Accounts Payable	9000
Equipment	45000	Jill Jay, Capital	25000
Insurance Expense	1800	Graphic Revenue	52100
Salaries Expense	30000	Consulting Revenue	6000
Supplies Expense	3700		
Advertising Expense	1900		
Rent Expense	1500		
Utilities Expense	1700		
	\$109100		\$109100

Additional data:

- (i) At June 30, \$1300 of supplies on hand
- (ii) Unexpired Insurance premium at June 30, \$450
- (iii) Depreciation on equipment for 6 month \$1000
- (iv) Graphic revenue earned but unbilled at June 30, totals \$2000

Instructions:

- (i) Prepare a statement of comprehensive income as on June 30, 2016 (for 6 months)
- (ii) Prepare owners' equity and
- (iii) Statement of financial position as on June 30, 2016.

Section – B

- 5. (a) Define Income Tax authority. (05)
(b) What is the basic objective to formulate administrative and judicial authority? (10)
(c) State the duties of deputy commissioner of taxes. (20)

 - 6. (a) What is meant by internal audit? (05)
(b) State the functions of internal audit. (10)
(c) State the scope of internal audit. (10)
(d) What are the demerits of internal audit? (10)

 - 7. (a) What do you mean by auditor's report? (05)
(b) State the characteristics of audit report. (10)
(c) Discuss the importance of audit report. (10)
(d) State the various types of audit report. (10)

 - 8. (a) Define operational auditing. (05)
(b) What are the objectives of operational audit? (10)
(c) What are the advantages of operational audit? (10)
(d) State the steps of oprational report. (10)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2016
URP 3161
(Housing and Real Estate Development)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define House and Housing. What is the difference between the concept of Housing and Real Estate? (10)
(b) "Adequate Housing is determined in part by social, economic, cultural, climatic, ecological and other factors". Explain the statement using ecological system theory. (12)
(c) In reference to the neo-liberal housing policy, who is responsible for providing housing for the poor? Explain your answer. (08)
(d) Explain the characteristics of a functional housing market. (05)
2. (a) "Urban areas are segregated geographical space". Explain this statement with reference to Dhaka city. (10)
(b) Write short notes on the following: (12)
 (i) Social housing
 (ii) Land banking
 (iii) Sites and services.
(c) Why did the government provision of housing for all failed in 1970s and 80s? Explain. (13)
3. (a) Why and how does international housing policy affect Bangladesh housing policy? (06)
(b) How did the housing policies change from 1970s to present? "The responsibility for housing forum became again government responsibility"-explain how? (15)
(c) What are your suggestions for solving urban housing problems? (07)
(d) Explain the major objectives of Bangladesh Fifth Five Year Plan (1997-2002) highlighting housing issues. (07)
4. (a) What is perfect market competition? What are the common non-market factors that influence land price in Bangladesh? (10)
(b) If you are assigned to estimate land price in Fulbarigate (beside the main road), list out the factors that will affect the price. Express the price function through hedonic assumptions. (15)
(c) What is Inclusive and Affordable housing in Singapore? What are its components?-explain in short detail. (10)



Section – B

5.
 - (a) Mention the fundamental ways to express the word real estate in empirical and professional field? List the three alternative uses with examples. (07)
 - (b) Exemplify the interactions of value determining sectors of real estate. (15)
 - (c) Illustrate the characteristics of a real estate market with necessary examples. (13)

6.
 - (a) Why is it that in a simple bid-rent model a person who commutes on foot will outbid one who commutes by car for the closest space to downtown? Explain the concepts with necessary figures. (15)
 - (b) What do you mean by investment in real estate? Explain the monetary and non-monetary benefits and costs for a real estate investment with examples. (12)
 - (c) Write down the ways to mitigate the investment cost uncertainty. (08)

7.
 - (a) Define economic base of a city. (05)
 - (b) Name the resources of a city. How do they contribute to urban growth? (12)
 - (c) Draw a bid-rent curve for multiple city centres and explain accordingly. (08)
 - (d) What is market research? Mention the key components of a good investment decision. (10)

8.
 - (a) Illustrate the factors which are responsible for affecting a real estate market. (15)
 - (b) How does the bid-rent curve relate to Burgess's Concentric Zone Model? (10)
 - (c) Calculate the NOI for a residential building from the following table where the annual income and expenditures are given. Also calculate the entry Cap rate i , when the purchase price is 900,000 tk. (10)

Unit	12
Rent/unit/month	1,200 tk.
Vacancy	10%
Property management fee	10%
Income from Laundry	5,000 tk.
Income from Parking	5,000 tk.
Income from Bill-board	2,500 tk.
Income from Advertisement	2,500 tk.
Insurance	5,000 tk.
Property taxes	14,000 tk.
Maintenance cost	14,000 tk.
Utilities expenses	20,000 tk.
Rent control expenses	1,000 tk.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2016
URP 3121
(Rural Development Planning-I)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Describe the necessity of Rural Development in the context of developing countries. (10)
- (b) "Urban Development is opposed to Rural Development", do you agree? Justify your answer. (10)
- (c) Discuss different problems of rural development. (07)
- (d) Why decentralization and participation is essential for rural development? (08)
2. (a) How per capita consumption expenditure is a better means of measuring rural development? (08)
- (b) Calculate nominal GDP and real GDP of a four goods economy in a small island nation "X". (10)

	Production		Price/ Unit	
	Yr 1 Q1	Yr 2 Q2	Yr 1 P1	Yr 2 P2
Goods A	55	102	0.50	0.40
Goods B	35	21	0.30	1.00
Goods C	40	52	0.70	0.90
Goods D	70	95	0.40	0.85

- (c) "GDP does not reflect increases in social welfare" – justify the statement with proper examples. (05)
- (d) Write a critique of the Human Development Index (HDI) as a measure of development. (12)
3. (a) "Poverty measure is a function of individual incomes and poverty line" – explain the statement (06)
- (b) What are the advantages and disadvantages of Poverty Gap (PG) Index? (10)
- (c) Determine (i) Poverty Gap (PG) Index and (ii) Squared Poverty Gap (SPG) Index for the following countries and explain the results. (12)

Expenditures for each individual in countries				
Country A	3500	5000	6300	7500
Country B	2500	4500	4700	3700
Country C	2600	4400	5000	6500

Assume the poverty line is 5000.

- (d) What are the policy implications of different poverty measures? (07)

4. (a) "Agriculture has a multitude of functions in the development process"- Explain them with appropriate examples. (13)
- (b) Write down the Strength and Threat of the agriculture sector under the analyzed SWOT in the National Agriculture Policy, 2013. (12)
- (c) Mention the opportunities and constraints considered for framing out the National Agriculture policy 1999 (10)

Section – B

5. (a) Mention different patterns of rural settlements. Use necessary sketches. (08)
- (b) Mention the characteristics of a rural community. (05)
- (c) Illustrate the procedures for determining present centrality and various hierarchical levels of rural settlements. Provide an example assuming necessary data. (16)
- (d) What is your understanding about the differences between urban centres and rural centres? (06)
6. (a) Give a brief overview of scattered settlements and gathered settlements with necessary sketches. (10)
- (b) Make an overview of different types of rural industries with appropriate examples. (15)
- (c) How does active campaign contribute to rural industrialization? Explain in short. (10)
7. (a) Illustrate the components of Comilla Model and their role in rural development. (15)
- (b) What were the causes of limited success of V-AID model for rural development? (10)
- (c) Show the trends of rural development programs in Bangladesh. (10)
8. (a) Illustrate different types of rural housing pattern with necessary sketches. (15)
- (b) Write short notes on the following policies for rural resources development: (15)
1. Price policy 2. Technology policy 3. Fiscal policy
- (c) What is your understanding about Swanirvar movement? (05)
